

CSCI 434 Final Project

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Abstract

This paper presents a machine learning-based approach to classifying web traffic originating from Amazon, YouTube, and Hulu without relying on IP address identification. As three of the most significant contributors to global internet traffic — with Amazon leading at 3.4% of global traffic as of early 2026, YouTube accounting for 12-16% in 2025, and Hulu maintaining a consistently active and growing user base — understanding the network behavior of these platforms carries meaningful economic and infrastructural implications. Beyond platform-specific analysis, this work examines the viability of generative AI as a tool for constructing predictive network traffic models on limited data, a question of growing relevance as Gen AI becomes increasingly embedded in industry development workflows. Network traffic was captured manually using Wireshark in 20-minute intervals across all three platforms simultaneously, with flows characterized by features including packet count, packet interval statistics, and packet length statistics. Data was cleaned to remove interference from advertisements and unintended applications, and classification models were developed with the assistance of ChatGPT and fine-tuned for accuracy. A hybrid ensemble model combining Support Vector Machines, Random Forest, and Gradient Boosting achieved the highest accuracy at 54%, surpassing the hypothesis threshold of 50%.

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1 Introduction

In this paper we work to understand the network traffic of Amazon, Youtube and Hulu. As of early 2026, Amazon is in the lead for the "Big Three" of online network traffic, even over Google and Microsoft, accounting for 3.4% of global internet traffic. This follows a growing trend of growth in both percentage of global traffic Amazon is responsible for, as well as Amazon's revenue, over the last 3 years. Further, as of 2025, Amazon was the most visited website in the world. Similarly, Youtube was responsible for between 12-16% of internet traffic in 2025. Hulu was responsible for just 2.1% of internet traffic; however, its user base is consistently active and growing. Together, these three platforms represent a substantial and economically significant portion of the modern

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internet's bandwidth consumption, making them ideal candidates for network traffic analysis.

Studying the network infrastructure of Amazon, Youtube, and Hulu, and specifically its interactions with other sites, is therefore important from an economic standpoint, as well as to maintain the integrity of our global internet infrastructure. Understanding how traffic from these platforms behaves at the packet level — without relying on IP address identification — offers a more flexible and broadly applicable approach to network analysis. As content delivery networks grow increasingly complex and traffic patterns evolve across platforms, developing tools to classify this traffic by behavioral characteristics alone becomes a meaningful technical challenge. Further, honing in on viable models to predictively measure and analyze network traffic allows cross-application of the models to analyze other sites.

Our dataset for this paper was composed of network traffic captured manually from Amazon, Youtube, and Hulu, with all three platforms running simultaneously on the same computer in 20-minute intervals. Data was cleaned for interference from ads redirecting traffic to other sites, and from there several models meant to predict where network traffic originated from were created using ChatGPT and fine-tuned for accuracy. A final model was created as a hybrid ensemble of Support Vector Machines, Random Forest Modeling, and Gradient Boosting, achieving an accuracy of 54% in predicting network traffic origin — surpassing the study's hypothesis threshold of 50%.

2 Proposed Method

This paper delivers a model to classify web traffic based on attributes other than an IP address.

2.1 Tool Preparation

Wireshark was used to capture data from three websites: Amazon, YouTube, and Hulu. QUIC protocol was disabled for ease of capture. All web traffic was considered. Claude AI was used to organize packets and flows and ChatGPT was used for the linear regression baseline, decision tree and SVM modeling. Websites and applications not meant to be included in the study such as TikTok and Snapchat, but were captured due to factors such as advertisements were dropped from the final dataset.

2.2 Data Preparation

Features considered for utilization from each flow were:

- **Packet Count:** The total number of packets per record, showing how much communication activity occurred. Higher counts suggest heavier traffic and can reveal usage patterns over time.

Train / Validation / Test Results							
Model	Train Acc	Train F1	Val Acc	Val F1	Test Acc	Test F1	Overall Gap
Logistic Regression	42.00%	37.81%	43.00%	34.92%	25.17%	27.10%	12.83%
Decision Tree	47.20%	43.83%	43.00%	37.39%	40.88%	39.03%	2.94%
SVM	55.50%	53.57%	57.00%	56.98%	41.87%	37.87%	15.47%
Random Forest	59.07%	59.18%	58.00%	58.70%	43.04%	41.62%	50.91%
Gradient Boosting	59.07%	59.18%	43.00%	42.82%	34.17%	33.14%	44.93%
Ensemble (Gradient+GB)	59.07%	59.18%	58.00%	58.70%	50.00%	48.52%	49.97%

* Overall Gap = Train Acc - Test Acc (Red = 0.00% = significant overfitting)
 * Green (GB) = (Area Under ROC Curve) / Real-World
 * Ensemble = each using of SVM + RF + GB (averaged class probabilities across all three models)
 * Gradient Boosting will output the Ensemble as F1 metric, more training data would reduce the gap

Figure 1: Table of data on key training data statistics

- **Average Packet Interval:** The mean time between consecutive packets, indicating how evenly traffic is spaced. Larger intervals point to bursty or intermittent traffic, while smaller ones suggest a consistent, steady stream.
- **Maximum Packet Interval:** The single longest gap between any two packets in a record, highlighting moments of inactivity or delay. This can reflect the responsiveness of the website or the nature of its traffic patterns.
- **Minimum Packet Interval:** The shortest time gap between any two consecutive packets, capturing how quickly data moves during active transmission periods. This is a useful indicator of network latency and overall site responsiveness.
- **Average Packet Length:** The mean size of packets across a record, giving a sense of how much data is typically carried per packet. This can hint at the kind of content being served — for example, small packets often suggest text-heavy pages, while larger ones point to media-rich content.
- **Maximum Packet Length:** The size of the largest packet in a record, representing the peak single-packet data transfer. Large values may indicate the presence of file downloads, images, or other rich media.
- **Minimum Packet Length:** The size of the smallest packet in a record, reflecting the lightest individual transmissions. Small minimums often correspond to brief control messages, acknowledgments, or lightweight requests.
- **Total Packet Length:** The total of the packet lengths. Indicated in summation the amount of network traffic.

Each flow was a datapoint of length 1 second.

2.3 Employed Method

The decision tree model was chosen as a baseline because of its simplicity in classification through bootstrap aggregation and majority voting to give a baseline result to compare against a strong model that may take longer to train. SVM was chosen because of its strength and commonality in classification problems using hyperplane to separate differing data as much as possible. Before being passed into the models, data was separated into training data, testing data and validation data using a 60/20/20 split.

Models were compared using accuracy in predicting the correct site and F-1 scores. A basic statistical analysis was performed on each website as well using pandas and NumPy. This study evaluated the hypothesis:

H1: Websites can be categorized with an accuracy of above 50% using an improved model of SVM and artificial intelligence.

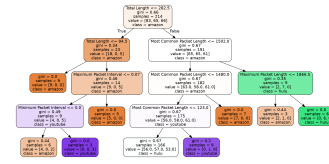


Figure 2: Decision tree based on the generated baseline decision tree model

3 Evaluation

3.1 Evaluation Metrics

We used two primary metrics to evaluate the models: Accuracy and Macro F1 score. These metrics provide a comprehensive view of performance across all classes. [See appendix for further details.] For each website, the packet mean and standard deviation were calculated throughout the entire capture.

3.2 Individual Website Data Profiles

A statistical analysis was done for each website to better understand reasons why each model may have struggled or thrived in certain areas.

Label	Size
Amazon	553
Hulu	548
YouTube	465
Total	1566

3.1.1 Amazon

Amazon had:

- a packet count $\mu = 318.94$ and a $\sigma = 944.66$ · a total length $\mu = 283204.27$ and a $\sigma = 887852.1$ · an average packet length $\mu = 511$ and a $\sigma = 376.61$, minimum packet length $\mu = 60.03$ and a $\sigma = 42.68$, maximum packet length $\mu = 2117.36$ and a $\sigma = 3101.67$
- an average packet interval $\mu = 0.0272$ and a $\sigma = 0.0591$, minimum packet interval $\mu = 0.0078$ and a $\sigma = 0.0438$, maximum packet interval $\mu = 0.1914$ and a $\sigma = 0.1658$
- a most common packet length of 1494 bytes

3.1.2 Hulu

Hulu had:

- a packet count $\mu = 811.96$ and a $\sigma = 2189.33$
- a total length $\mu = 785490.64$ and a $\sigma = 2144896.50$
- an average packet length $\mu = 448.30$ and a $\sigma = 458.33$, minimum packet length $\mu = 61.07$ and a $\sigma = 31.55$, maximum packet length $\mu = 1651.11$ and a $\sigma = 4504.74$
- an average packet interval $\mu = 0.0383$ and a $\sigma = 0.0661$, minimum packet interval $\mu = 0.0099$ and a $\sigma = 0.0387$, maximum packet interval $\mu = 0.1968$ and a $\sigma = 0.2033$
- a most common packet length of 1510 bytes

3.1.3 YouTube

YouTube had:

- a packet count $\mu = 310.21$ and a $\sigma = 846.57$
- a total length $\mu = 300,858.24$ and a $\sigma = 841091.85$
- an average packet length $\mu = 391.60$ and a $\sigma = 401.74$, minimum packet length $\mu = 68.44$ and a $\sigma = 65.37$, maximum packet length $\mu = 1960.67$ and a $\sigma = 3133.43$

Train / Validation / Test Results							
Model	Train Acc	Train F1	Val Acc	Val F1	Test Acc	Test F1	Credit Gap
Logistic Regression	42.00%	37.00%	43.00%	34.00%	28.17%	27.10%	12.80%
Decision Tree	47.20%	43.00%	43.00%	37.00%	39.94%	39.94%	4.14%
SVM	55.14%	53.00%	52.00%	48.00%	41.47%	37.80%	13.47%
Random Forest	59.07%	58.00%	58.00%	56.70%	43.00%	41.60%	56.01%
Gradient Boosting	59.07%	58.00%	43.00%	42.82%	54.17%	53.14%	44.80%
Ensemble (GB+RF+GB)	59.07%	58.00%	58.00%	58.70%	50.00%	48.50%	49.80%

* Credit Gap = Train Acc - Test Acc; Test Acc: Real < 0.200 = significant overfitting
 * Clean (GB) = Area 30-40% / Real < 100%
 * Ensemble = self voting of SVM + RF + GB (averaged class probabilities across all three models)
 * Gradient Boosting will edges the Ensemble as F1 metrics more training data would make the gap

Figure 3: Table with calculated accuracies for the several models generated by the GenAI

Network Traffic Classification — Model Performance Summary							
Model	Test Accuracy	F1 Score	Precision	Recall	Shannon	Shannon	Shannon
Logistic Regression	28.17%	27.10%	36.16%	27.00%	27.00%	45.00%	50.00%
Decision Tree	43.00%	39.94%	40.34%	40.00%	44.00%	43.00%	47.00%
SVM	41.47%	37.80%	39.31%	39.12%	39.24%	37.80%	50.14%
Random Forest	43.00%	41.60%	40.07%	41.00%	41.70%	39.80%	50.07%
Gradient Boosting	54.17%	53.14%	54.07%	53.00%	53.10%	43.00%	50.07%
Ensemble (GB+RF+GB)	50.00%	48.50%	49.00%	48.70%	48.70%	39.80%	50.07%

Figure 4: Table of the raw accuracies from each of the models generated for training and for performance. Further includes information on training data statistics.

• an average packet interval $\mu = 0.0427$ and a $\sigma = 0.0952$ minimum packet interval $\mu = 0.0184$ and a $\sigma = 0.0812$, maximum packet interval $\mu = 0.1644$ and a $\sigma = 0.2074$

• a most common packet length of 55 bytes

3.1.4 Other Statistical Test

The Cohen D and the Mann-Whitney test were performed as well, and showed that the differences between each feature of each website are significant but the effect size was small for each category, with most features falling in between the ± 0.1 -0.3 range.

3.3 Baseline Model Analysis

Using the decision tree model as a baseline, we found that the model had the ability to predict website traffic with a 62.74% accuracy and had a precision rate of 68.34% and recall rate of 60.91%, giving it an F1 score of 60.18%. Per class, the model was able to have a precision rate of 62.6% for Amazon traffic and recall rate of 73.9%, giving it an F1 score of 67.8%. Hulu had a precision rate of 58.6% of the time with a recall rate of 80.9%, giving it an F1 score of 67.9%. YouTube had the highest precision rate of 83.9% and the lowest recall rate of 28%, giving it the lowest F1-score of 41.9%. The training data had a 60.28% accuracy and a F1-score of 56.31%. The validation set had an accuracy of 60.06% and a F1-score of 53.44%.

3.4 Improved SVM Model Analysis

The improved SVM model had an overall improvement from the baseline model, recording an accuracy of 69.75% with precision rate of 69.15% and a recall rate of 68.96%, giving a weighted F1-score of 69.48%. Per class, the model had a precision rate of 68.6% for Amazon, with a recall rate of 74.8%, giving a F1-score of 71.6%. Hulu had a precision rate of 75.9% and a recall rate of 77.3%, giving a F1-score of 76.6%. YouTube had the lowest results, with the model recording a precision rate of 63% and a recall rate of 54.8%, giving a F1-score of 58.6%. For each dataset in the split, the train accuracy was 97.23% and an F1-score of 97.11%. The validation set had an accuracy of 66.13% and a F1-score of 65.53%.

Confusion Matrices — Test Set											
Logistic Regression						Decision Tree					
Actual \ Predicted	Amazon	Hulu	YouTube	Amazon	Hulu	YouTube	Actual \ Predicted	Amazon	Hulu	YouTube	
Amazon	11	0	0	11	0	0	Amazon	10	0	0	
Hulu	0	0	0	0	0	0	Hulu	2	18	0	
YouTube	0	0	0	0	0	0	YouTube	0	0	0	

SVM						Random Forest					
Actual \ Predicted	Amazon	Hulu	YouTube	Amazon	Hulu	YouTube	Actual \ Predicted	Amazon	Hulu	YouTube	
Amazon	10	0	0	10	0	0	Amazon	10	0	0	
Hulu	0	0	0	0	0	0	Hulu	2	18	0	
YouTube	0	0	0	0	0	0	YouTube	0	0	0	

Figure 5: Table of confusion matrices results from each model.

Per-Class Performance — Test Set											
Model	Class	Precision	Recall	F1-Score	Support	% OT Baseline F1					
Logistic Regression	Amazon	33.3%	10.4%	41.0%	21	5.8pp					
	Hulu	33.3%	27.7%	30.5%	22	-3.9pp					
Logistic Regression	YouTube	18.2%	18.2%	18.2%	22	-15.0pp					
	Amazon	88.2%	50.1%	45.9%	28	Baseline					
Decision Tree	Hulu	38.0%	81.8%	50.0%	22	Baseline					
	YouTube	44.4%	55.6%	50.0%	22	Baseline					
SVM	Amazon	40.0%	57.0%	55.1%	28	+16.0pp					
	Hulu	35.7%	22.7%	27.8%	22	-23.0pp					
SVM	YouTube	30.0%	27.3%	28.6%	22	+1.0pp					
Random Forest	Amazon	50.0%	57.1%	53.3%	28	+8.0pp					
	Hulu	47.2%	31.8%	35.9%	22	+4.0pp					
Random Forest	YouTube	34.8%	54.5%	45.6%	22	+19.0pp					
Gradient Boosting	Amazon	64.3%	64.3%	64.3%	28	+20.0pp					
	Hulu	52.0%	40.9%	46.2%	22	+9.0pp					
Gradient Boosting	YouTube	44.4%	54.5%	49.5%	22	+20.0pp					
Ensemble (GB+RF+GB)	Amazon	58.2%	64.3%	60.0%	28	+16.0pp					
	Hulu	55.0%	34.4%	42.1%	22	+7.0pp					
Ensemble (GB+RF+GB)	YouTube	47.7%	40.9%	45.3%	22	+17.0pp					

Figure 6: Per-class performance statistics on each individual model.

4 Discussion & Future Work

The improved SVM model showed the expected results and outperformed the hypothesis across all web traffic.

4.1 Decision Tree

The baseline decision tree model struggled with organizing and telling the difference between YouTube and Hulu and learned to have an automatic Hulu prediction when receiving streaming service traffic, which is shown in the low recall of YouTube. Both are adaptive bitrate streaming services, but this confusion is also proven by quantitatively by looking at conducting a t-test between Hulu and YouTube on Average Packet Interval, which gives a p-value of 0.3945, which is not statistically significant, which makes it hard to distinguish between Hulu and YouTube, versus a t-test between Amazon and Youtube which gives a p-value of 0.0016 which is significant. Looking at the mean σ for packet intervals, 91.4% of Hulu flows fall within YouTube's. The combination of both of these stats plus the overlap of the other features such as total length, packet count and more falling into at least one standard deviation from the mean error between classifying the web traffic is understandable.

4.2 Improved SVM

The improved SVM model worked well because of the use of Gradient Boosting, which was able to find the small differences between YouTube and Hulu through multiple iterations as well as combining multiple features to find a wider disparity than just looking at one. For example, if there is a flow that exists in the overlap with average packet interval between YouTube and Hulu, gradient boosting will combine the flow feature with other features, and if those features end up on the lower side it will more likely predict YouTube and if it is higher than it might choose Hulu. Using these combined features, the 200 sequential trees in Gradient Boosting learn these features. Through this sequential error correction in addition to Random Foresting, which helps by using majority voting across

the 200 trees with random features, any flow has a good chance of being correct because soft voting will choose whoever feels most confident about a particular flow.

4.3 Feature Ceiling and Limitations

P-values between Hulu and YouTube across both models were not significant (average packet interval $p=0.3945$). This shows the problem is not the model or the amount of data, but just the type of features and the amount of them. The improved SVM model was better at picking up on the small differences, but because the difference is very weak, reaching 100% would be impossible. Future research should look at adding more features such as burst size, burst count, throughput, TLS record size and more in addition to the nine from the study.

Due to how hierarchy of the data, for example in average packet length, any traffic can be clearly defined as Amazon or not Amazon, but after looking at packet length the model then has to use the rest of the features to distinguish between Hulu and YouTube which have high overlap.

4.4 Future Work

Due to the limitations of this study, we propose the following as avenues for the future of this project:

- **Dataset Expansion:** Broadening the dataset to cover a wider range of websites and traffic records, while also capturing behavior over extended time periods, would strengthen the model's training foundation and improve its ability to generalize across diverse scenarios and perform reliably under varying conditions.
- **Advanced Models:** Investigating more sophisticated machine learning approaches — such as deep learning architectures or ensemble methods — could boost classification accuracy by better capturing the complex, non-linear relationships present in network traffic data.
- **Practical Applications:** Beyond academic work, future work should prioritize deploying the model in real-world network monitoring, management, and security contexts. Partnering with industry stakeholders would help stress-test and refine the model across a variety of operational environments, demonstrating its broader utility.

5 Conclusion

In this project, we developed and evaluated a model for monitoring and classifying network traffic by its originating website. Using a dataset collected from diverse websites, we employed three models: Logistic Regression, Decision Tree and an Improved SVM. Through comprehensive evaluation, the Improved SVM model was found to outperform the others, achieving the highest accuracy and Macro F1 score. This work demonstrates the feasibility of machine learning models, particularly, for handling complex relationships in network traffic data. Accurate classification of network traffic patterns contributes significantly to efficient network management, security, and performance optimization. The main insights include: 1) The Improved SVM model's ensemble approach provides robustness and superior performance, particularly in handling diverse traffic patterns. 2) The project's results indicate the importance of dataset

size and diversity in model training, highlighting the need for expansion in future work. 3) The project's visualizations reveal how each model's predictions align with actual values, providing further insights into their effectiveness. This project serves as a foundation for future developments in network traffic analysis, with practical implications for real-world systems and applications. By integrating machine learning models into network monitoring, we can enhance performance, security, and management.

6 Appendix

- (1) Accuracy is the proportion of correct predictions (both true positives and true negatives) among the total number of cases examined [?].

$$Accuracy = \frac{TN + TP}{TN + FP + TP + FN}$$

where TP = True positive; FP = False positive; TN = True negative; FN = False negative.

- (2) F1-Score computes how many times a model made a correct prediction across the entire dataset [?].

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$F1\ Score = 2 * \frac{Precision * Recall}{Precision + Recall}$$

where TP = True positive; FP = False positive; TN = True negative; FN = False negative.

A References

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